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Fracturing Bottomhole Pressure Comparative Study: Automated Data-Driven Algorithm vs. Downhole Gauge vs. Manual Calibration

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Abstract

Accurate bottomhole pressure (BHP) estimation is pivotal for the success of hydraulic fracturing treatments, affecting both operational safety and stimulation effectiveness. In the absence of downhole gauges, engineers typically rely on surface pressure readings combined with frictional and hydrostatic components. However, traditional workflows that use lab-derived fluid friction tables and generalized proppant multipliers fail to capture the well-specific dynamic behavior during treatment. In this study, we benchmark three BHP estimation approaches: (1) a default method using static lookup tables from the flow loop tests, (2) a manually calibrated method using downhole gauge data and engineering judgment, and (3) an automated, data-driven methodology based on real-time ISIP detection and machine learning-predicted friction components. The evaluation spans three fluid systems, crosslinked gel, linear gel, and slickwater. The dataset consists of vertical wells instrumented with pressure gauges. Results show that the automated approach outperforms both traditional methods in accuracy and consistency. For crosslinked fluids, it reduced RMSE and MAPE errors by 35–55% compared to manual and default workflows. In slickwater treatments, the performance of all methods was comparable; however, the automated system demonstrated better capture of near-wellbore complexity without human intervention. The study also explores abnormal transient pressure phenomena post shut-in and highlights the critical dependence of ISIP quality on accurate water-hammer signal analysis. Overall, the automated approach enables scalable, objective, and field-calibrated BHP estimation that eliminates the need for extensive human subjectivity, manual labor, which at best averages the errors from multiple datasets through interpolation, and at worst, is erroneous due to faulty methods and assumptions. The automated approach treats the well for its individual observations which can be unique.

Introduction

Bottomhole pressure (BHP), defined as the pressure exerted at the top of perforations during pumping, is a fundamental control parameter for hydraulic fracturing design and execution. It governs critical processes such as fracture initiation, fracture propagation extent & direction, tip screenout and packing, proppant transport, and near-wellbore behavior. For decades, engineers have used surface treating pressure as a ground truth measurement appended with hydrostatic and frictional losses estimation to calculate BHP in

wells lacking pressure gauges. However, these corrections introduce uncertainty, especially in the friction components, which are affected by fluid type, proppant loading, rate, temperature, perforation conditions, and near-wellbore complexity.

During treatment design, a reliable BHP estimation ensures that pressures exceed breakdown and insitu stress thresholds while avoiding near-wellbore bridging or screenout. In operations, it is critical for optimizing the pump schedules, particularly the proppant ramps, fluid design and diversion strategies. Inaccurate BHP predictions can lead to (a) lack of control over fracture geometry, (b) underused proppant placement potential of the generated fracture, (c) suboptimal fracture diversion, (d) near-wellbore screenouts, etc. Moreover, the value of installing a downhole gauge lies in its ability to eliminate uncertainty in tubular friction but since it is installed few 100s of ft shallower than perforations or production interval, it provides no direct information about perforation losses or tortuosity between casing and formation.

Traditional workflows use dynamic flow loop-derived fluid friction curves and static proppant multipliers. These values, though simple to implement, do not scale up and do not reflect the variability in field operations. Manual calibration with downhole gauge data partially addresses this gap but suffers from subjectivity and some art to build the full curve. The single point (rate) frictions acquired with gauges vary significantly even in the same tubing/casing size and fluid type and it leads to interpolation errors when used to build a universal friction curve which dilutes the well-specific behavior. Additionally, interpreting gauge data to adjust friction curves is labor-intensive.

To address these limitations, [Ugarte et al. \(2025\)](#) introduced a novel automated workflow that uses ISIP from each stage to dynamically calibrate fluid friction curves. The method uses oscillation-based water hammer interpretation to extract accurate ISIP from surface data. Once ISIP is known, the fluid friction gradient is computed from surface pressure by removing hydrostatic and other relatively well-known pressure components. This approach, even though valid for single stage wells (using the data from nearby wells), is particularly powerful in multiple stage wells, where this approach builds a dynamic curve reflecting real well conditions. Proppant friction is incorporated via machine learning models trained on time-series operational data from multiple wells across fluid systems. Perforation friction is assumed negligible in the later stages due to erosion but is corrected using pump-down pressure (PDP) or surface diagnostic tests (SDTs) where needed. Near-wellbore tortuosity is excluded from the ISIP-derived friction but can be included using field diagnostics or PDP signatures when appropriate.

This paper builds on that methodology (significantly detailed in [Ugarte et al. 2025](#)) and compares it to conventional workflows using a field dataset containing gauge-equipped vertical wells treated with crosslinked, linear gel, and slickwater systems. The goal is to assess the fidelity of BHP estimation under three contrasting workflows: (1) the default static approach, (2) a manually calibrated approach using expert interpretation of gauge data, and (3) the automated friction extraction and ML-driven method. For benchmarking, downhole gauge BHP is treated as ground truth, and comparisons are made using RMSE, MAPE, and expert-ranked visual scoring.

This work thus addresses two major questions: How accurate are common BHP estimation methods? And what scalable alternatives exist when gauge data is unavailable or limited? The findings show that automated, data-driven friction estimation provides both accuracy and scalability, marking another step forward for digital fracturing workflows.

Automated Approach Overview

The proposed methodology observed in [Fig. 1](#) enhances bottomhole pressure estimation by reducing uncertainties associated with three key friction components: fluid friction, proppant friction, and perforation friction. This data-driven approach presented in previous studies by [Ugarte et al. \(2025\)](#) enables a more accurate pressure estimation to enable higher efficiency of the treatment for fracture design and execution optimization.

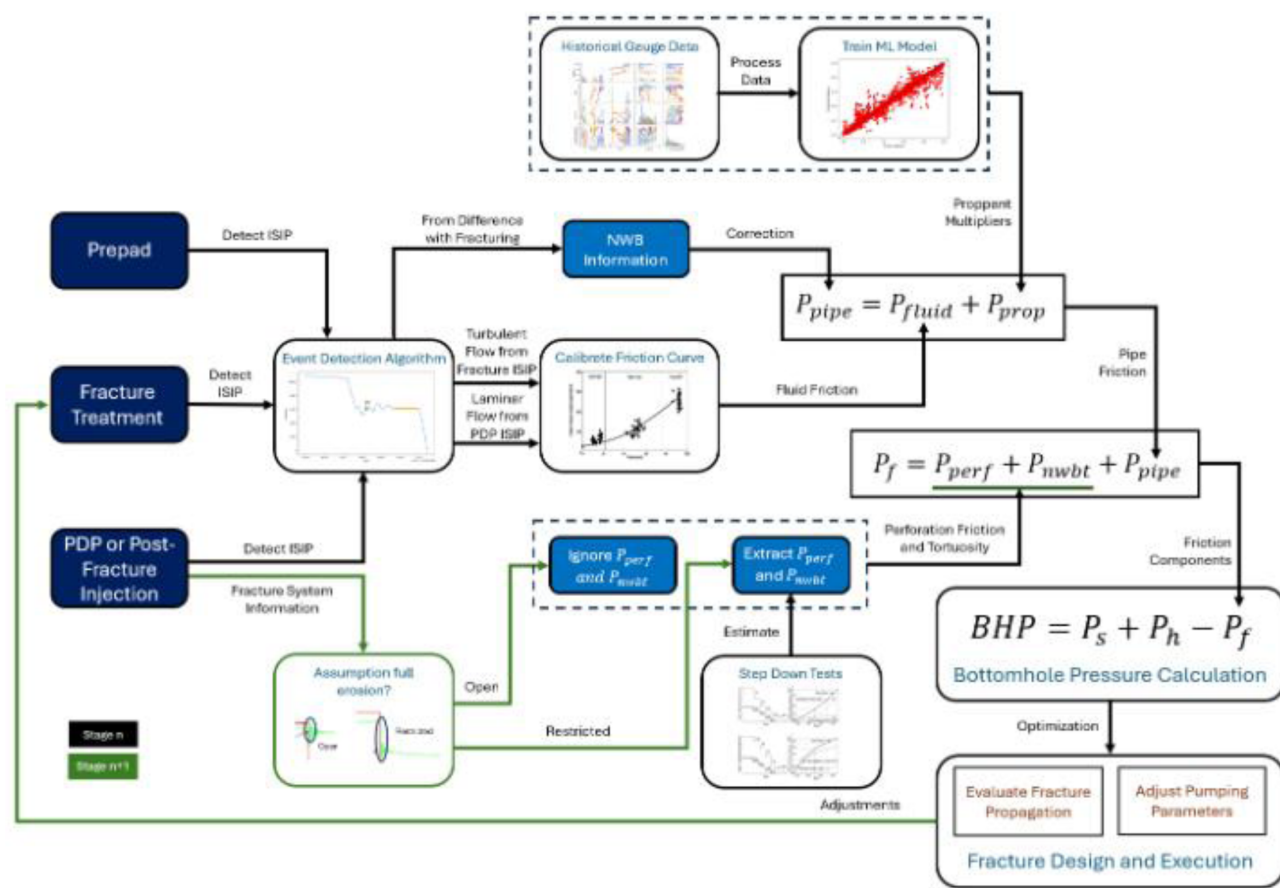
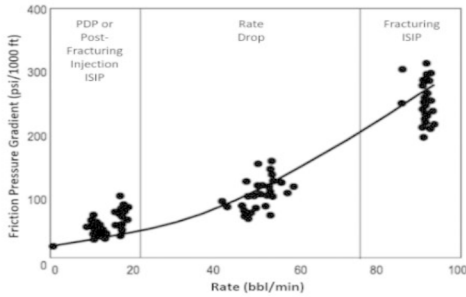
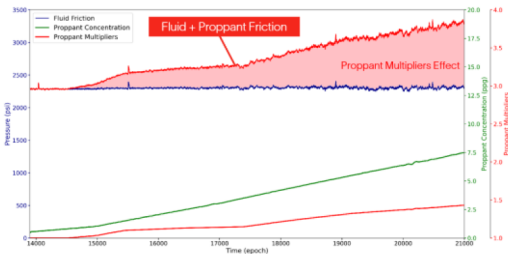
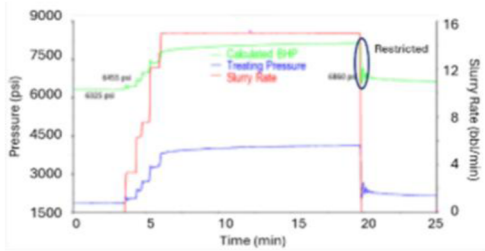
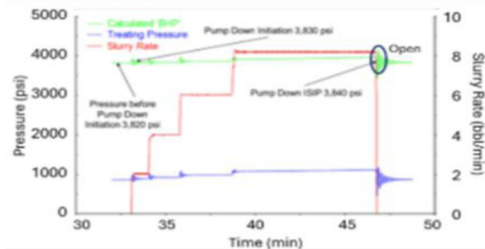


Figure 1—BHP calculation workflow using different components (fluid friction, proppant multipliers, perforation entry loss, NWBT) and their respective sources (Ugarte et al. 2025).

Table 1 provides a consolidated summary of the four major components that influence bottomhole pressure estimation: fluid friction, proppant friction, perforation entry friction, and near-wellbore tortuosity.

Table 1—Summary table for various BHP calculation components in the new automated approach.

Component	Description	Automated Workflow Method	Visual Representation	Considerations & Limitations
Fluid Friction	Friction from clean fluid (no proppant) flowing through tubulars	Data-driven ISIP-based friction curves using both PDP and fracturing rates. ISIP extracted via automated detection of post-shutdown water hammer oscillations	 Figure 2—Friction pressure gradient versus rate curve building process(after Mu et al. 2022)	- Replaces static lab-based flow loop curves with real-time curves - Relies on clean fluid ISIP accuracy - Curve accuracy increases with number of stages per well or representative reference offset wells
Proppant Friction	Additional friction from slurry due to proppant type, concentration, and size	ML-predicted proppant multipliers using time-series gauge-labeled data and operational parameters (rate, pressure, etc.)	 Figure 3—Proppant multipliers dataset formulation logic showing the isolated proppant friction contribution from gauge data.	- Avoids discontinuous pressure steps seen in static tables - Captures dynamic operational behavior - Model trained with multiple wells across fluid types
Perforation Entry Friction	Pressure loss from fluid entry and flow through small, restrictive perforations	Negligible at end of stage due to perforation erosion; PDP pressure drop used to correct if system is restricted		- Assumption valid only in open systems - Restricted systems require PDP analysis or SDT corrections
Near-Wellbore Tortuosity (NWBt)	Pressure drop due to complex flow paths between perforation and formation in the near-wellbore region	Mostly excluded in late-stage ISIP drop; SDTs recommended to characterize where necessary		- Decays during treatment but not fully removed - Requires 3-rate SDT tests to decompose tortuosity vs. perforation effects - Correction used only if PDP indicates restricted system

BHP Calibration Process

To assess the effectiveness of the proposed methodology, a comparative case study was conducted using a dataset comprised of multiple vertical wells equipped with gauge data. These wells had various fluid families

and were used in comparison with different methodologies. The wells used in this study were instrumented with downhole pressure gauges and surface sensors, enabling continuous measurement of injection pressure, rate, proppant concentration, and temperature.

Design of the Gauge Calibration Study

To calibrate the friction coefficient across various completion types and fluid systems, bottomhole gauge (BHG) data from seven wells was analyzed. These wells underwent a total of 17 stimulation treatments. Four distinct fluid types were used throughout these treatments: brine, linear gel, slickwater with friction reducer, and crosslinked gel. The proppant concentration in each treatment varied significantly, ranging from 0.5 to 8 pounds per gallon added (PPA), with high-strength proppant utilized in all cases.

All the wells involved in this study were exploratory and completed with permanent downhole gauges, primarily installed for extended well testing purposes. During fracturing operations, real-time bottomhole (BH) data was utilized to optimize fracture treatments, enabling more informed, on-the-fly decisions based on actual downhole behavior.

The BH gauge data also provided direct measurements of temperature and friction profiles during and after the fracturing treatments, especially when different fluid types were used. This data proved invaluable for understanding how various fluids interact with the wellbore and formation under dynamic conditions.

Typically, the well completions employed two tubing sizes: 5.5 inches and 4.5 inches, as illustrated in Fig. 5 below. Two gauges were installed in the 4.5-inch tubing—one serving as a backup and the other enabling accurate hydrostatic pressure measurements between the gauges during flowback. These gauges were spaced approximately 100 meters apart, allowing for precise friction measurements within the 4.5-inch tubing.

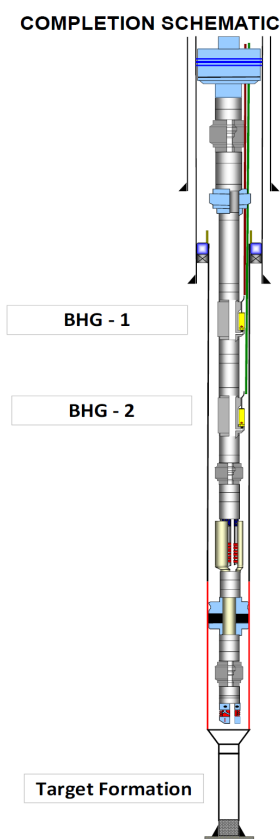


Figure 5—Typical completion with two gauge used for the calculation.

Friction in the 5.5-inch tubing was calculated indirectly, using the data from the 4.5-inch tubing and adjusting for the respective tubing lengths. This setup provided a reliable method for estimating friction losses across different tubing sizes and contributed to a more accurate calibration of the friction coefficient

Actual Measurement of the Tubing Friction

The measurement of the friction in 4.5 in tubing utilized the following equation 1:

$$P_f = P_{ug} + (P_{hyd-ug} - P_{hyd-lg}) - P_{lg} \quad (1)$$

where P_f is the friction pressure; P_{ug} is the pressure reading at upper gauge; P_{hyd-ug} is the hydrostatic pressure at upper gauge; P_{hyd-lg} is the hydrostatic pressure at lower gauge; P_{lg} is the pressure reading at lower gauge. Based on the measurement and calculation from 6 wells using the above equation, the following result (Fig. 6) obtained from the gauge raw data for brine, linear gel, slickwater and crosslink fluid

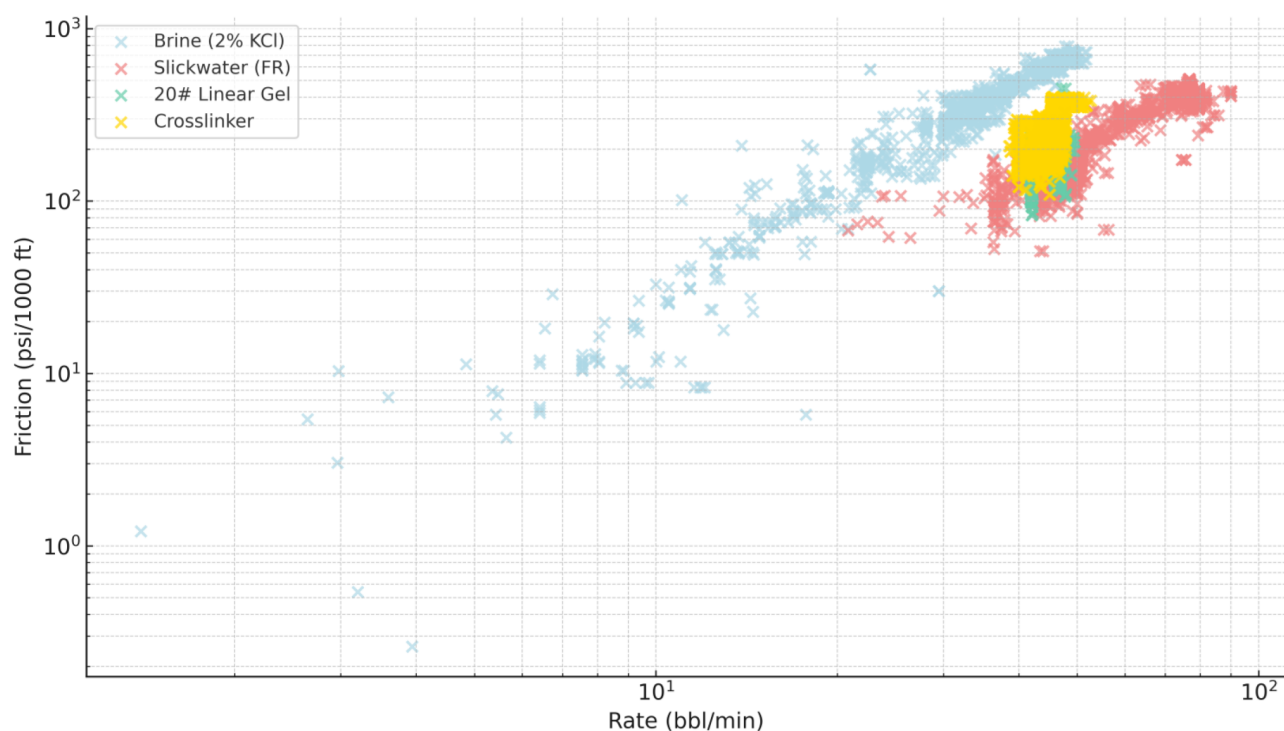


Figure 6—Friction gradient (psi/1000-ft) data for brine, slickwater, linear gel, and crosslinked gel in 4.5-inch tubing with 3.74-inch ID

Once the friction data of 4.5 in obtained then the friction of 5.5 in tubing friction calculated using below equation:

$$P_{f-5.5} = (P_s + P_{hyd-lg} - P_{lg}) - P_{f-4.5} \quad (2)$$

where $P_{f-4.5}$ is the friction Pressure in 4.5 in; P_{ug} is the pressure reading upper gauge; P_{hyd-ug} is the hydrostatic pressure upper gauge; P_{hyd-lg} is the hydrostatic pressure lower gauge; P_{lg} is the pressure reading lower gauge. The friction results for both the 4.5-inch and 5.5-inch tubing were then tested independently in a different well to validate the findings. These results have been simplified and summarized in Fig. 7, which provides a comparative overview of friction behavior across both tubing sizes under similar operating conditions.

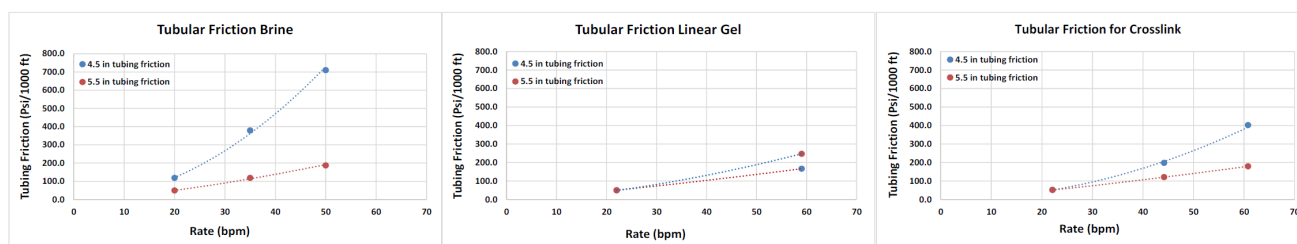


Figure 7—Curve approximations from all the friction measurement data for brine, linear gel, and crosslinked gel in 4.5-inch and 5.5-inch tubing

The final component of the friction analysis is the proppant friction multiplier, which is calculated by subtracting the friction of the PAD fluid from the friction observed at each proppant concentration. This approach isolates the additional friction caused specifically by the presence of proppant.

Fig. 8 illustrates how varying proppant concentrations affect friction in crosslinked fluid. Unfortunately, the measurements were limited to flow rates between 44 and 48 barrels per minute (bpm), which correspond to the most stable rates observed during the pumping operations. Despite this limitation, the data provides valuable insight into how proppant loading influences friction behavior under typical operating conditions.

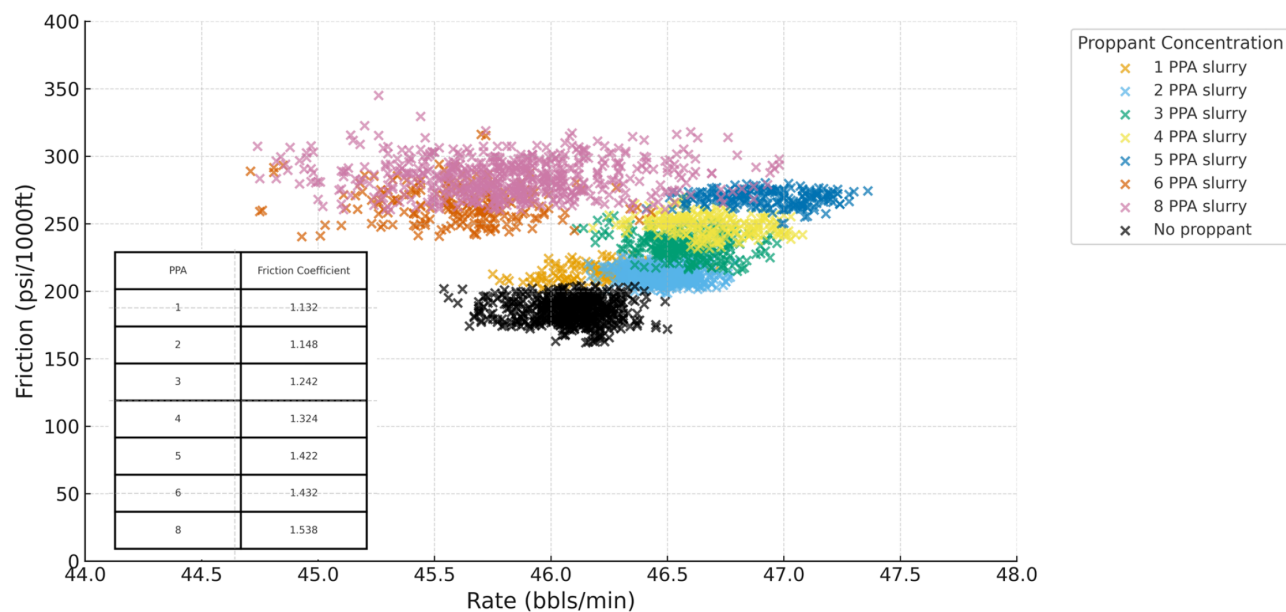


Figure 8—Friction gradient (psi/1000-ft) data in different proppant concentration using crosslink fluid in 4.5-inch tubing with 3.74-inch ID

Comparative BHP Results

The default approach utilized standard industry best practices using static friction curves derived from laboratory testing. However, discrepancies between laboratory and field conditions often lead to deviations between expected and observed values during operations. The manually calibrated method employed downhole gauge data and domain expert insights to adjust fluid friction gradients based on the observed behavior of previous similar wells in the area. The approach is based on the subjectivity of the domain expert fit to the extracted information from the gauge data. Perforation friction and NWBT effects were accounted in the adjusted calibrated friction curve. Proppant multiplier effects were approximated using engineer-defined corrections from the observed behavior to manually build a proppant multipliers table. For the sake of comparison, the same proppant multiplier table was used for both the default and manually calibrated method. Lastly, the proposed methodology employed automated ISIP detection for dynamic

well based friction estimation and machine learning-based proppant multiplier prediction. The approach is to continuously self-calibrate when multiple stages are present reducing reliance on static assumptions or subjective decisions. In this case, the same perforation friction and NWBT tortuosity corrections were applied to the default approach and the automated methods. These corrections require additional tests to obtain information regarding the actual behavior observed, in this case that available information was averaged across the available data and applied to account for this phenomenon in the BHP calculation.

For each treatment, BHP estimates were generated using all three methods. The results were benchmarked against gauge-measured BHP values using three performance metrics: expert ranked qualitative score, root mean square error (RMSE) and mean absolute percentage error (MAPE). The expert ranked qualitative score was a ranked assigned comparison based on visual subject matter expertise deeming 1 to be the best and 3 to be the worst. RMSE measures the average magnitude of errors in pressure units. MAPE provides insight into the percentage deviation of estimated value relative to the actual gauge values, giving a comprehensive evaluation of the difference between the calculated with the actual.

The wells from the study were conventional vertical wells, for this reason the data was single stage treatment. This represents a limitation of the study as the extracted data used to generate the friction curve was from the end stage for each case. Nevertheless, this still shows the validity of the methodology. Plus, with more stages to generate the friction curve the results become statistically accurate by reducing uncertainty and possible erroneous measurements. A case study with multiple stages in a single well can be observed in [Ugarte et al. \(2025\)](#).

For crosslink fluid, [Fig. 9](#) presents the BHP profiles for six different wells using all three methods alongside the actual BHP measured using downhole gauges. The default method consistently overestimates the friction pressure generating an underestimated BHP across the entirety of each stage. This is due to its inability to adapt to each well condition. The use of static controlled laboratory measurements for friction values per fluid family can vary from one location to another. The expert calibrated method shows an improved alignment with gauge measurements compared to the default method. However, similar to the default method it suffers from the inability to adapt to each well condition because it generalizes multiple wells information into a single friction curve. Although it has been derived from gauge data it can be observed that it can underestimate and overestimate the BHP depending on the well. Furthermore, the process of gathering the gauge data is expensive and the process to generate the friction curve from the measured data is labor intensive and biased to the domain expert performing the fit. Both default and calibrated method utilized a proppant multiplier table that generate a ladder step behavior when switching from one multiplier range to another as observed in the [Fig. 6](#). The proposed automated method demonstrates the closest match to the measured gauge BHP across the majority of treatments. This is due to the ability to adjust the friction curve to the specific operation. In addition, the ML model used to predict the proppant multiplier generate better estimation avoiding the step changes within the multipliers range. NWB and tortuosity information is needed for all cases regardless of the method used to improve the accuracy of the estimation.

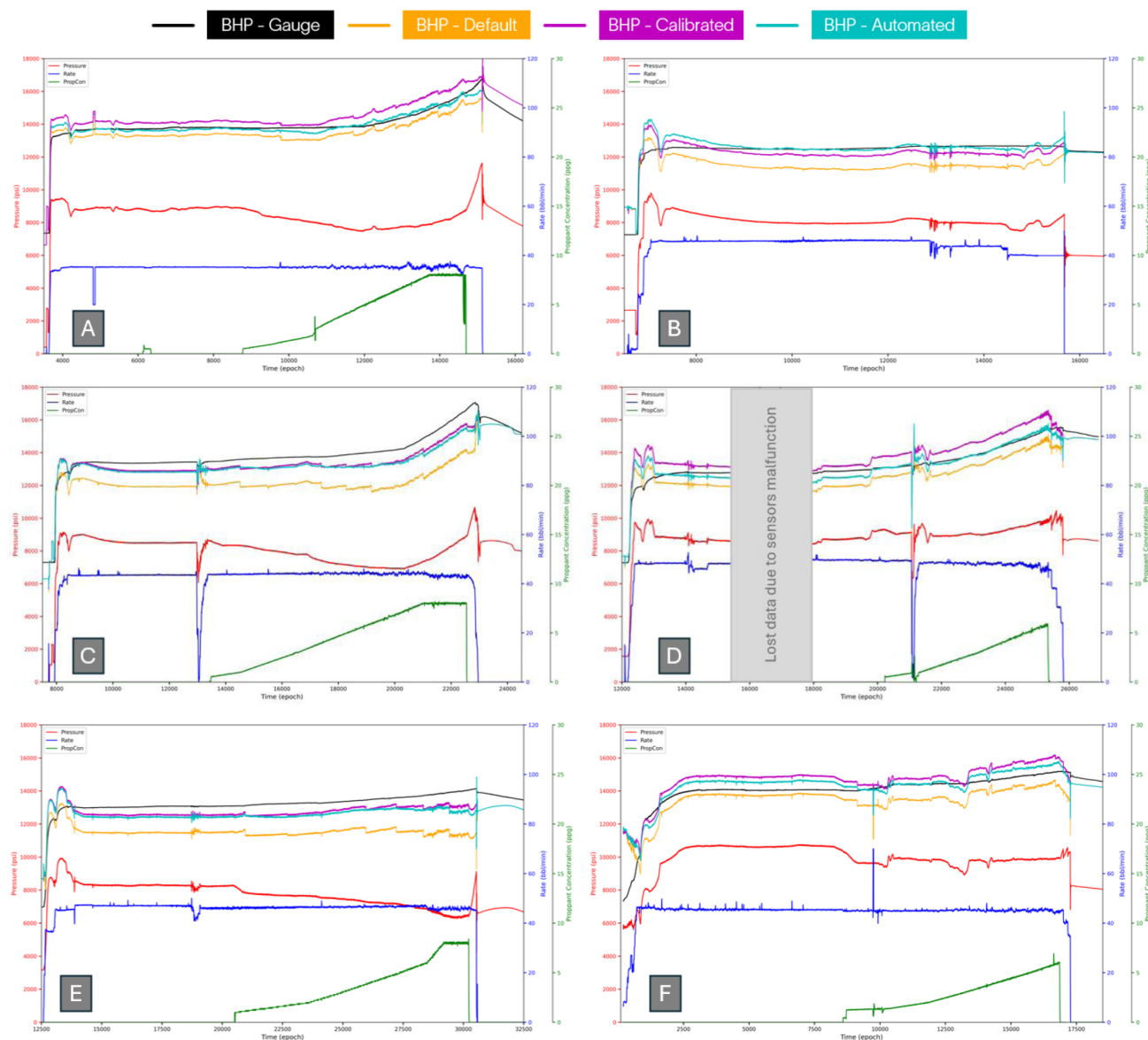


Figure 9—Crosslink fluid friction comparison for Wells A to F, comparing the default, calibrated and automated methods against BHP measured using downhole gauge data.

The friction curves for the default and calibrated methods can be observed in Fig. 10. Notably, the automatically generated friction points from each well form a bounded region, highlighting how each well produces a unique curve, particularly at higher injection rates. This emphasizes that friction behavior is well-specific. It is important to mention that the calibrated curve is already accounting for NWB perforation and tortuosity effects, whereas the default and automated use a correction to the friction based on a default average value across all the observed wells. It is important to highlight that low-rate data points are absent from this case study; in this instance, the friction curves were extrapolated to cover the lower-rate regime. Nonetheless, collecting friction data across a broader range of rates would provide more accurate friction curve construction and deeper insight into the interaction between fluid flow, perforation geometry, and near-wellbore complexity.

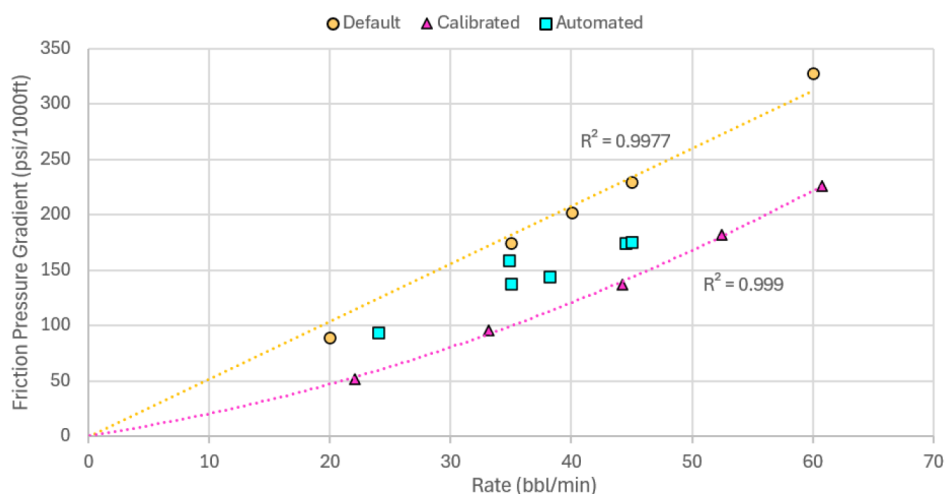


Figure 10—Friction curves for crosslink fluid, default and calibrated methods with each well extracted friction pressure gradient from the automated method.

The average performance results can be observed on Fig. 11. Quantitative metrics clearly demonstrate the superior performance of the automated method, which achieved an average RMSE of 781 psi and MAPE of 4.5%. These results show strong alignment with the qualitative expert rankings, where the automated method was consistently rated the most accurate. The average results for the calibrated method ranking second in the qualitative score were 1203 psi and 7.9% for rmse and mape respectively. In contrast, the default method exhibited the highest error, with an average RMSE of 1728 psi and MAPE of 12%, and ranked lowest in the expert evaluation. Notably, the automated method's performance held an excellent performance showing approximately 35% lower error compared to the calibrated and 55% relative to the default for crosslinked fluids highlighting its robustness and practical applicability.

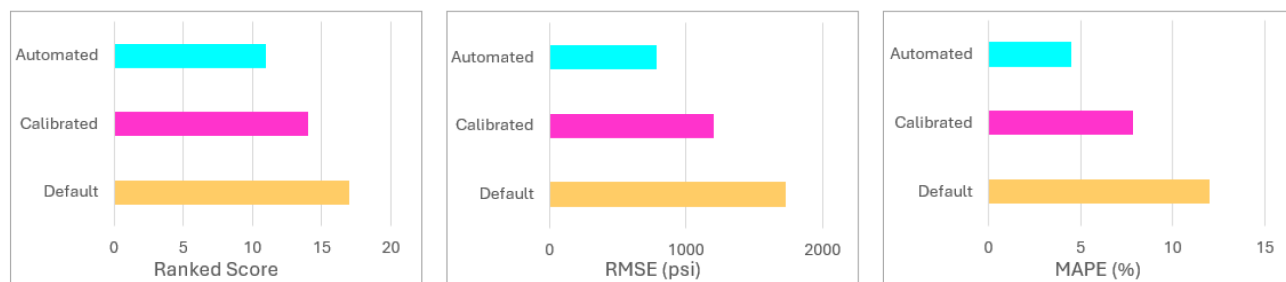


Figure 11—BHP average results for Crosslink fluid comparing the default, calibrated and proposed methods against the recorded value from downhole gauges.

However, it is important to note that the accuracy of the automated estimation is highly dependent on the quality of the ISIP measurements. The more stages available in a single well, the more statistically robust the friction estimation becomes. For example, in Wells C and E we observe some atypical behavior. In well-E, when pumps shutdown, the BHP gauge shows normal transient behavior, but the surface pressure increases for a short period of time. This is a pipe only behavior occurring downstream of the gauge due to either surface transducers, temperature effects, etc. Moreover, in Well-C, both gauge BHP and surface pressure show the same abnormal behavior indicating the pressure anomaly is occurring upstream of the gauge depth. This could be due to, (a) loss of communication between wellbore and fracture due to the complete near-wellbore proppant pack (bridging or screenout like); (b) fracture tip extension at shutdown followed by elastic rebound due to rapid closure, which can increase local pressure downhole and propagate that to the surface; (c) elastic energy release from the near-wellbore tortuous path. These phenomena affect the ISIP generated at the end of the stage overestimating friction generating the BHP values to the lower.

While in the other wells the behavior is as expected generating a better estimation. It shows how sensitively the ISIP estimation can be dependent on the water hammer quality.

For linear gel, the data included only three wells (Fig. 12) where two of these had operating issues, leading to near-wellbore screenouts. These wells generate invalid signals post stages that do not allow for proper friction curve extraction. If extracted, the values would massively underestimate the friction. This is because once the fracture pack transitions from tip packing to near-wellbore packing and bridging, the pressure following that is completely masked as the wellbore is isolated from the fracture. The ISIPs in these screenout treatments are not representative. However, well A did not experience any issue during or after the treatment, as such it generated a friction curve that when used on the well generated good BHP profile. For analysis purposes, the same friction curve was used on the other two wells. It can be observed in Figure 8 that the results were accurate and comparable with the other methods. However, when looking at the evaluation metrics in Fig. 13 we can observe that on average the results were RMSE of 627 psi and MAPE of 4.2%. Compared to 582 psi and 3.6% for default method and 504 psi and 2.6% for the calibrated method for RMSE and MAPE respectively. Qualitatively with the ranked score the calibrated was significantly better while the other two methods were almost identical. Average metrics were the lowest, showing the advantage of well-to-well unique curves and how minor variations can exist between them even in similar completions. Nonetheless, all three methods exhibited strong overall performance for linear gel fluids, with a maximum difference of only ~ 120 psi in average RMSE.

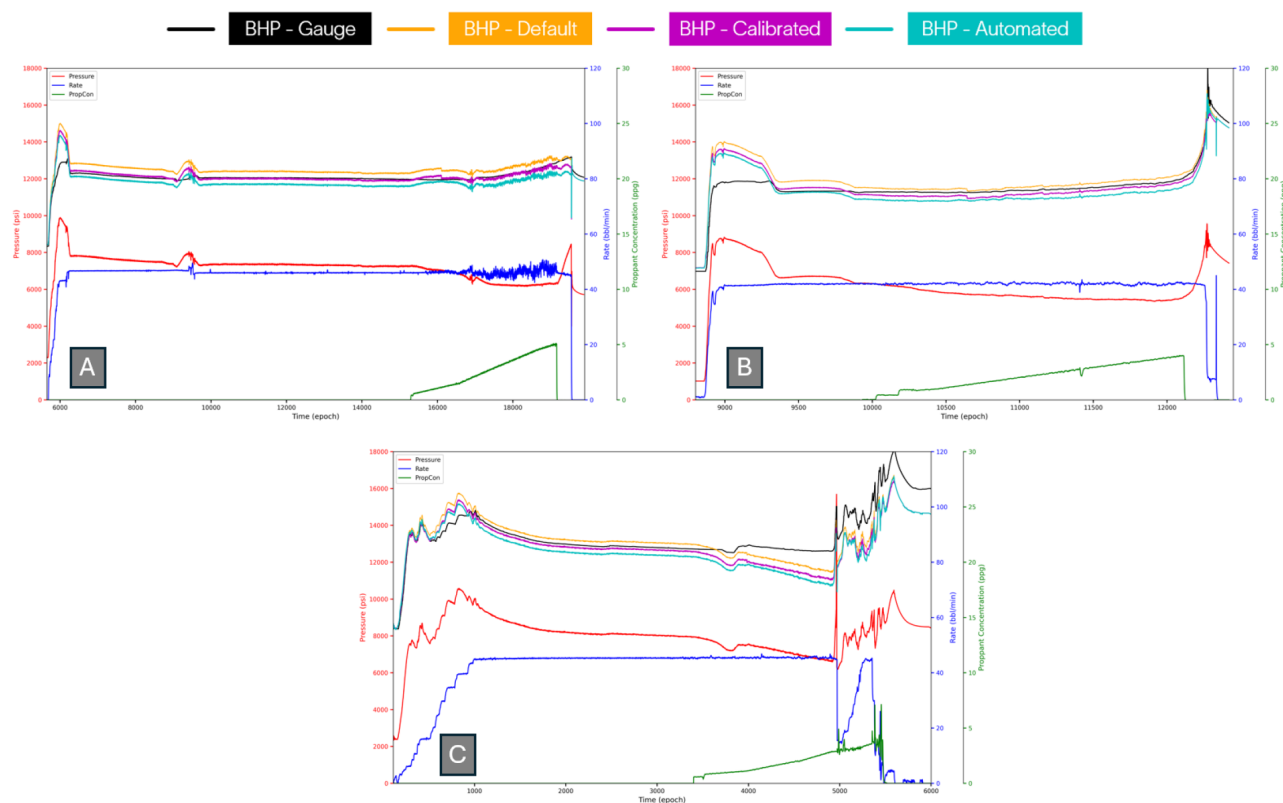


Figure 12—Linear gel fluid friction comparison for Wells A to C, comparing the default, calibrated and automated methods against BHP measured using downhole gauge data.

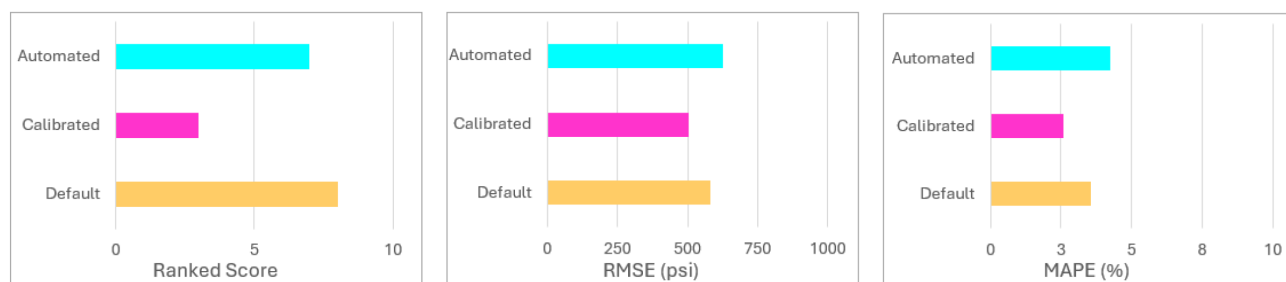


Figure 13—BHP average results for Linear Gel fluid comparing the default, calibrated and proposed methods against the recorded value from downhole gauges.

The five wells with slickwater fluid can be observed on [Fig. 14](#). Among them, wells A and B exhibited different near-wellbore (NWB) perforation effects and tortuosity compared to Wells C through E, despite using the same fluid type. This variation highlights the sensitivity of bottomhole pressure (BHP) estimation to near-wellbore conditions. The magnitude of these affected is evident in the gauge measurements, where restricted pressure behavior reveals flow resistance due to NWB complexity. Slickwater fluids are lower viscosity systems that create narrower fractures with higher lengths but limited proppant transport efficiency. Causing perforation erosion to be lower compared to more viscous fluids. As such, proppant concentrations as observed in [Fig. 14](#) do not exceed 3 ppg. To account for the near wellbore perforation and tortuosity separate correction factors were applied to the friction curves for Wells A–B and Wells C–E.

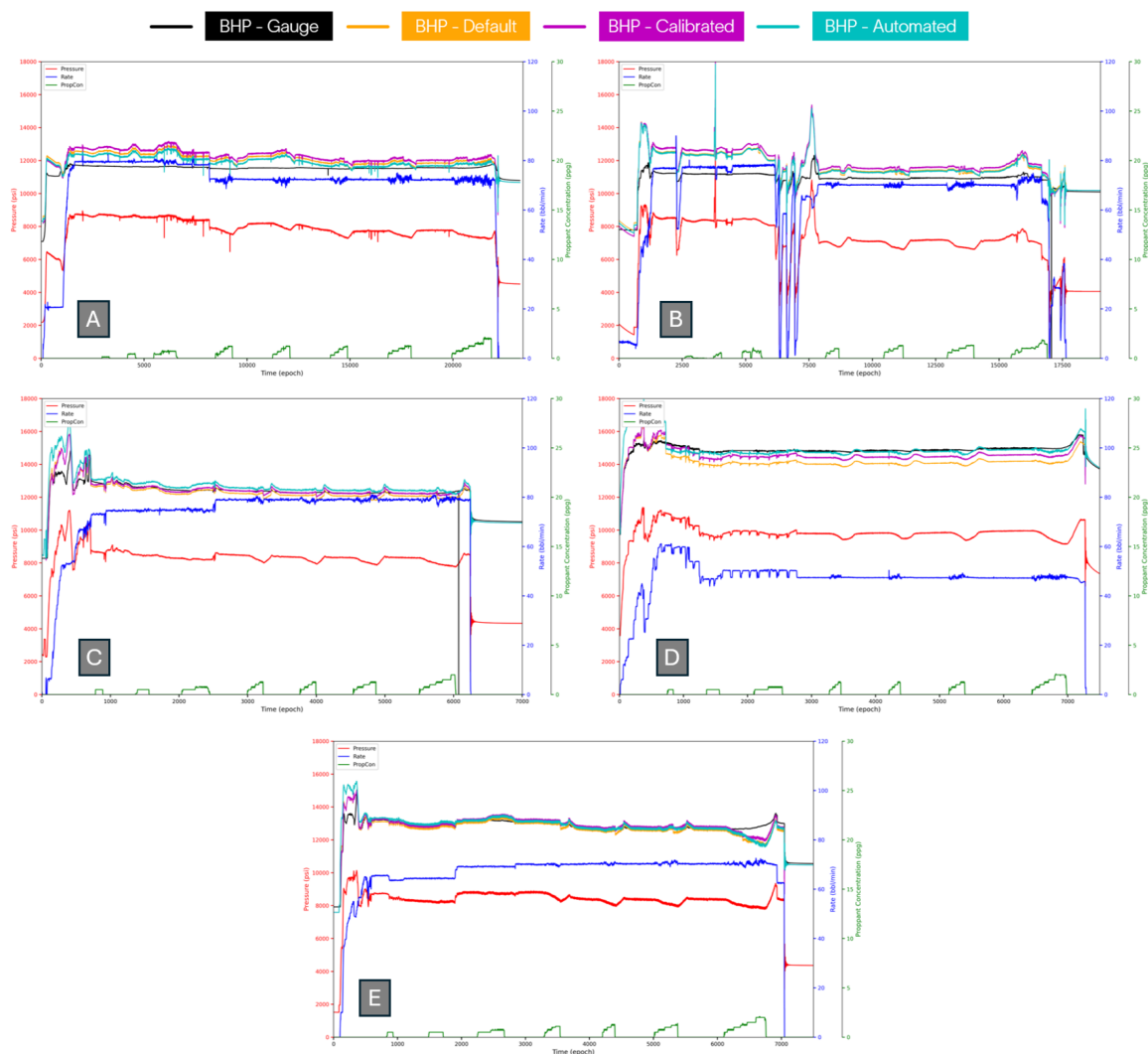


Figure 14—Slickwater fluid Wells A to E comparing the default, calibrated and automated methods against BHP measured using downhole gauge data.

The comparative performance results are summarized in Fig. 15. The default method showed to have a average RMSE of 570 psi and MAPE of 3.8%; the calibrated method to have RMSE of 585 psi and MAPE of 4.1%, while the automated method with 553 psi and 3.2% for RMSE and MAPE respectively. In the qualitative expert ranking, the automated method was rated highest, while the default method was ranked lowest. Despite these rankings, the quantitative differences among all three methods were minimal, with a maximum RMSE deviation of only ~32 psi, indicating relatively consistent estimation performance under slickwater conditions.

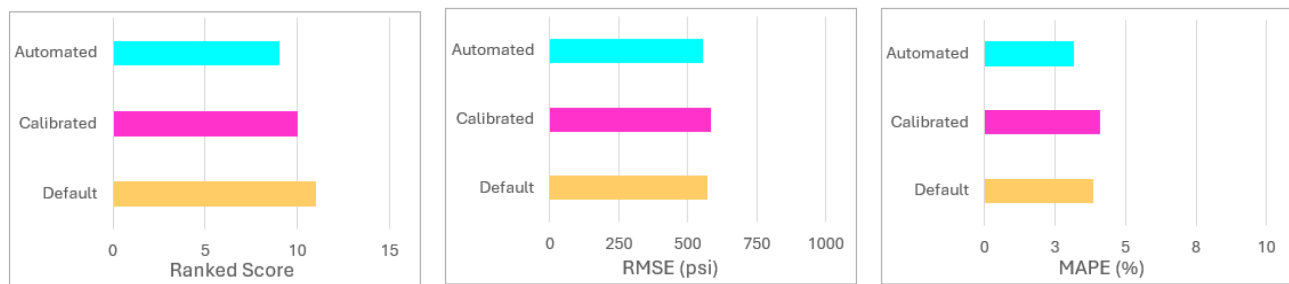


Figure 15—BHP average results for Slickwater fluid comparing the default, calibrated and proposed methods against the recorded value from downhole gauges.

The friction curves for the slickwater default and calibrated methods can be observed on Fig. 16. The automated friction points generated for each well form two distinct regions: Wells A and B, which exhibited lower NWB perforation and tortuosity friction, aligned closely with the default friction curve; whereas Wells C through E, which exhibited higher NWB resistance, fell into a separate higher-friction region. The calibrated method applied corrections to bridge this gap. The automated approach shows very distinct behavior by showing the near wellbore components embedded into fluid friction. During gauge calibration, these corrections can be made manually but if overlooked, the calibration wrongly averages the fluid friction with the near-wellbore loss components. For the automated approach, we have implemented two actionable techniques: (1) use the results from SDT and other diagnostic and plug it into the BH module; (2) if PDP or minifalloff is happening, those pressures can also be analyzed to extract these. Hence, the overall difference in BHP results between the curves remained relatively small, reflecting the inherently lower frictional variability in slickwater treatments compared to more viscous fluid systems. The automated approach treats each pressure trend individually without averaging the errors similar to the calibrated approach which uses multiple datasets and interpolates within the data involving some art along with science.

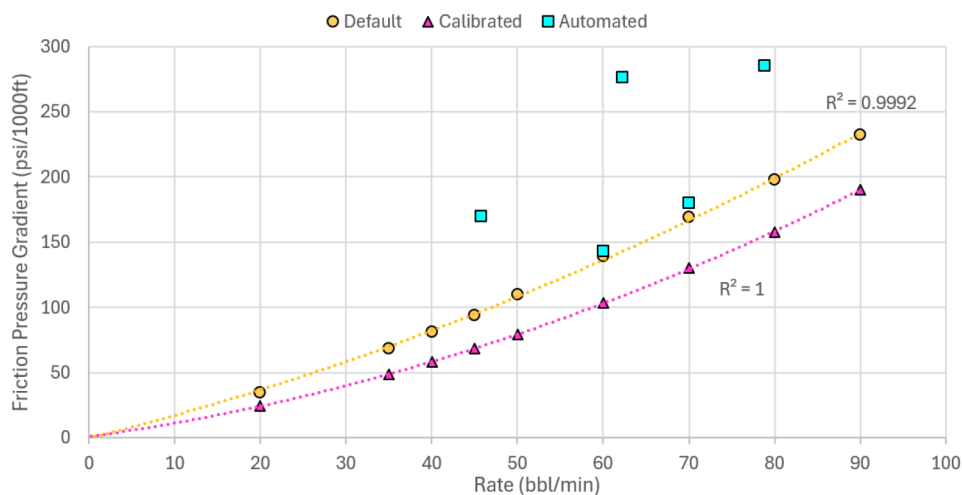


Figure 16—Friction Curves for slickwater, default and calibrated methods with each well extracted friction pressure gradient from the automated method.

The results underscore the limitations of static and manual methods, especially when applied to increasingly complex hydraulic fracturing treatments involving viscous fluids and higher proppant concentrations/volumes. Default method exhibited the lowest performance across all methods showing the existing gap between controlled laboratory conditions and field operations. While the calibration method yields reasonable accuracy, it is inherently subjective and hard to scale across multiple wells. By contrast, the

automated method exhibited the best results overall offering a scalable, objective, and accurate framework for estimating BHP, enabling improved treatment evaluation and design.

Discussion and Way Forward

When estimating BHP during stimulation, it is often safer to slightly overestimate rather than underestimate, especially for high-pressure, low error margin reservoirs. Overestimation may lead to conservative decisions such as early screenout detection or under-designing the proppant concentration ramp, but these are manageable risks compared to the operational hazards of underestimated BHP which can lead to approaching boundaries of the compliance envelope. This can be caused by premature screenout, unrecognized tip extension, or fracture migration into undesirable zones. Therefore, designing models that err conservatively helps remain on the safe side of the fracturing envelope and we have used that approach by modularizing the near wellbore component.

In the comparative study, the automated approach yielded superior results for crosslinked gel systems. A key reason is that in viscous fluid systems, the actual fluid friction component dominates the total pressure drop, while near-wellbore losses and perforation friction are relatively lower or more predictable. The higher viscosity provides enough energy to erode perforations as well as tortuous near-wellbore pathways. In contrast, for slickwater and linear gels, low viscosity and failure to get rid of the near-wellbore friction leads to relatively lower tubular friction, and the influence of near-wellbore effects becomes disproportionately larger. These components, if not properly characterized, can skew the ISIP-derived friction values, leading to either under- or over-estimated BHP. However, as shown in the automated approach, such deviations can be corrected by either isolating the NWB pressure drop using diagnostic tests or adjusting the ISIP-based curve across multiple stages for better robustness.

Future work could explore integrating DFIT or diagnostic mini-fracs to decompose ISIP more precisely into near-wellbore and fracture propagation components. Another area of opportunity is embedding real-time correction modules that flag ISIP anomalies post shutdown due to pressure rebounds, perforation packoff, or fracture extension. With the increasing adoption of digital workflows, such enhancements will be key to building adaptive and modular BHP estimation tools that scale across various basins and treatment types.

Summary and Conclusions

This study evaluated three methods for estimating bottomhole pressure (BHP) during hydraulic fracturing and benchmarked their performance against downhole gauge data across multiple vertical wells. The compared approaches included (1) the default method using static friction tables (from flowloop tests) and lookup proppant multipliers, (2) a manually calibrated method leveraging domain expert interpretations of gauge data, and (3) a fully automated, data-driven workflow using stage-wise ISIP extraction and ML-predicted proppant friction.

Across all three fluid systems (crosslinked gel, linear gel, and slickwater), the automated method demonstrated the highest accuracy and robustness. For crosslinked systems, it reduced RMSE and MAPE errors by more than 35–55% compared to traditional workflows. It dynamically adapted to the unique operational signature of each well, eliminating reliance on subjective calibration or generalized friction assumptions. While the default and manual methods showed acceptable performance in slickwater and linear gel treatments, their reliance on averaged friction curves or pre-defined correction tables led to mismatches when field conditions diverged towards more edge conditions.

Overall, the results validate the automated workflow as a scalable and technically rigorous solution for estimating BHP in wells without gauges. Its ability to auto-calibrate friction curves using surface data alone offers substantial operational advantages, particularly in large-scale, multi-stage treatment campaigns where

gauge coverage is sparse or impractical. The findings underscore the role of data-driven models in advancing stimulation diagnostics and reducing reliance on manual engineering judgment.

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